

TECH IN INDIC LANGUAGES

THE LANGUAGE BARRIER IS KEEPING MILLIONS OF INDIANS FROM ENJOYING THE BENEFITS OF TECHNOLOGY, AND NOW CUTTING EDGE TECH IS BEING USED TO OVERCOME THAT BARRIER.

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All around the world, most people use technology in their native languages. However, in India, the population remains underserved because of a number of reasons, even though 6 Indian languages are in the top 20 of the World, when considering the number of speakers. More people speak Panjabi as compared to say Korean, Italian or French. There are more than sixty languages in India with over 100,000 speakers, and 30 with over a million. The 29 states in India use 22 scheduled or constitutionally approved languages. The sheer linguistic diversity is staggering. Only about twelve percent of the population speaks English, and for most of them, it is a second language.

MACHINE TRANSLATION

One of the most elementary services for Indic language users is that of translation. Several other services, including applications, intelligent keyboards, voice assistants and chatbots can, at times, depend on good machine translation to work their magic. Translation services can mess up in ridiculous ways, but over the past decade, there has been a sustained effort



to bring machine translation closer to human levels. These efforts are primarily driven by advances in AI. Older generations of translators used what is known as phrase-based translations, where sentences were translated into bits and pieces. Newer methods replicate the ways people understand connections between words and context within neural networks, and then use those for translating entire sentences. Typically, the machine learning models are trained on pairs of translations, say legal contracts, news reports, or politi-

cal documents which are available in Chinese and English, and translated by humans. These data sets contain millions of parallel sentences in the two languages. Now these translation models are based on just feeding in as much data as possible. The more data the neural net is trained on, the more accurate the translations are. However, many of the Indic languages are low resource. Such datasets are not available for all the possible language pairs in the world. Even if the resources are available, they may not be in the